Agent factory

Hand work to cloud coding agents and keep every change reviewable. Five tools and a team harness repo, for a team of six running about sixty agent sessions.

hangar RUN	Runs the agents
flightdeck JUDGE	Judges the runs
Tower GOVERN	Governs risky actions
Harness DEFINE	Defined in the team harness repo
ai-dlc SUPPORTS DEFINE	Authors the workflow plugin
Spaceport CAPABILITY	Serves data to agents

— THE PROBLEM

Sixty agents at once breaks the way a team works

LIVE SESSIONS

~60

Six engineers, about ten streams each. Nobody can watch them all.

REVIEW

1 gate

Every change still needs a human. Reviewers become the bottleneck.

HARNESS CHANGES

0 evidence

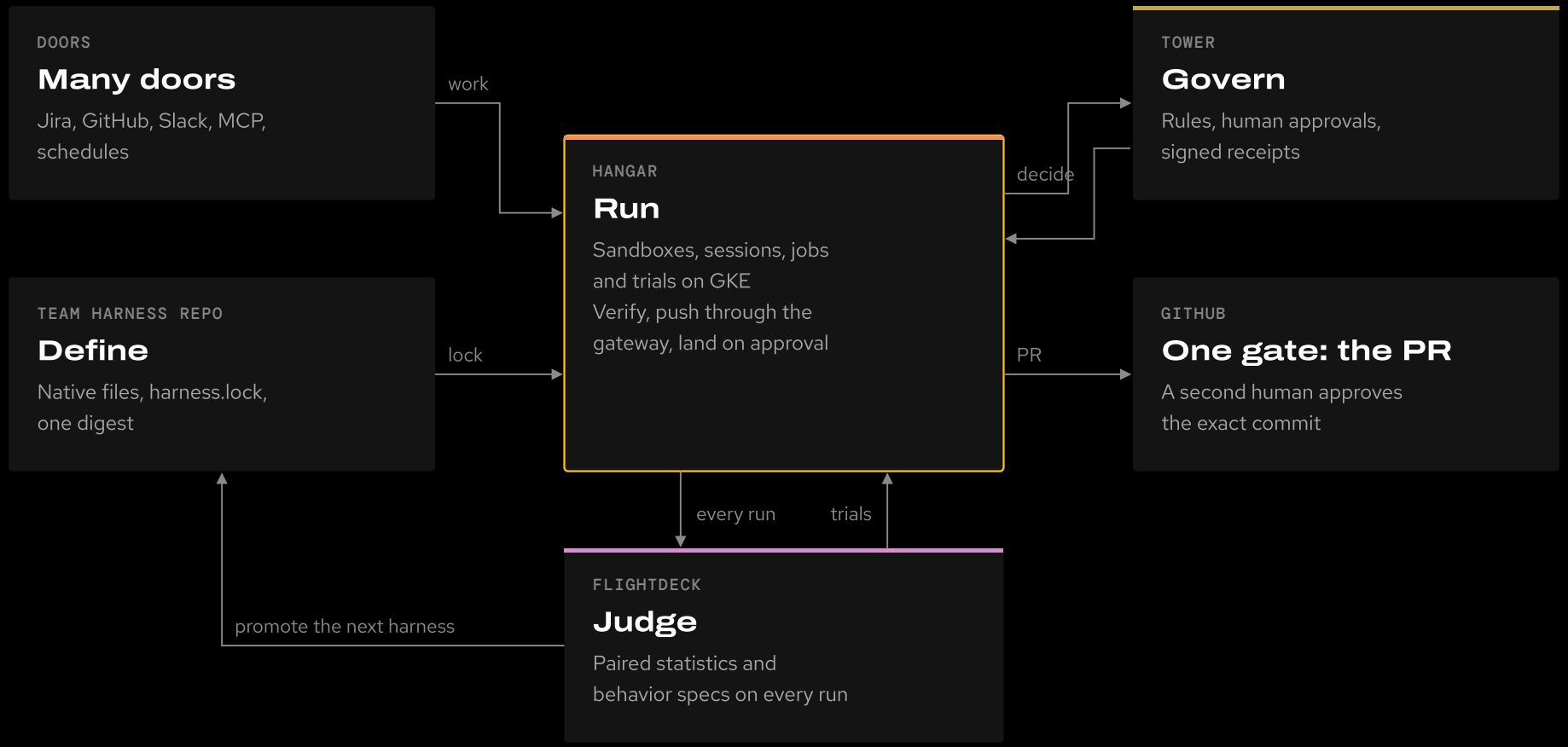
A new skill or prompt is judged by impressions, not paired runs.

UNTRUSTED INPUT

Every ticket

Ticket text and web pages can steer an agent toward a push.

Four planes, one loop, one gate



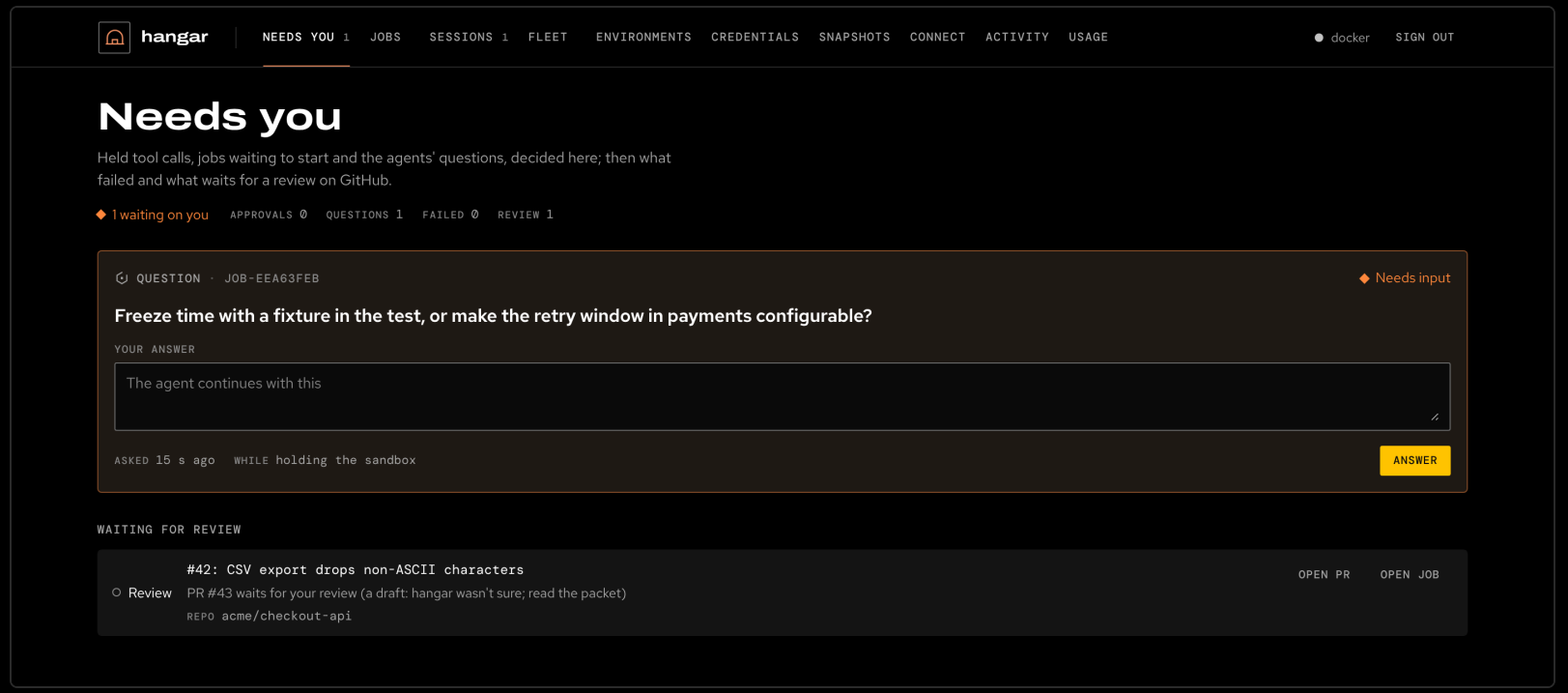
A Jira ticket to a merged PR

01	**Jira**	An engineer moves the ticket to “Ready for agent”	QUEUED
02	**hangar**	Ticket text is untrusted, so the job starts tainted	ADMITTED
03	**Sandbox**	The agent works with a pinned harness; every step is traced	RUNNING
04	**Gateway**	hangar verifies, then pushes one commit as the GitHub App	DELIVERED
05	**Reviewer**	A draft PR waits for a second human to approve the exact commit	NEEDS APPROVAL
06	**Outbox**	The merged run is written once, for flightdeck to judge	CAPTURED

Ten agents. One needs you.

Needs you is the console's first tab: held tool calls and agents' questions to decide here, then what failed and what waits for a review on GitHub. Everything running stays grey.

- A question is answered in place; the agent continues with the answer.
- A deny needs a reason. It goes on the receipt and back to the agent.
- The clear state is the hero: **Nothing is waiting on you.**



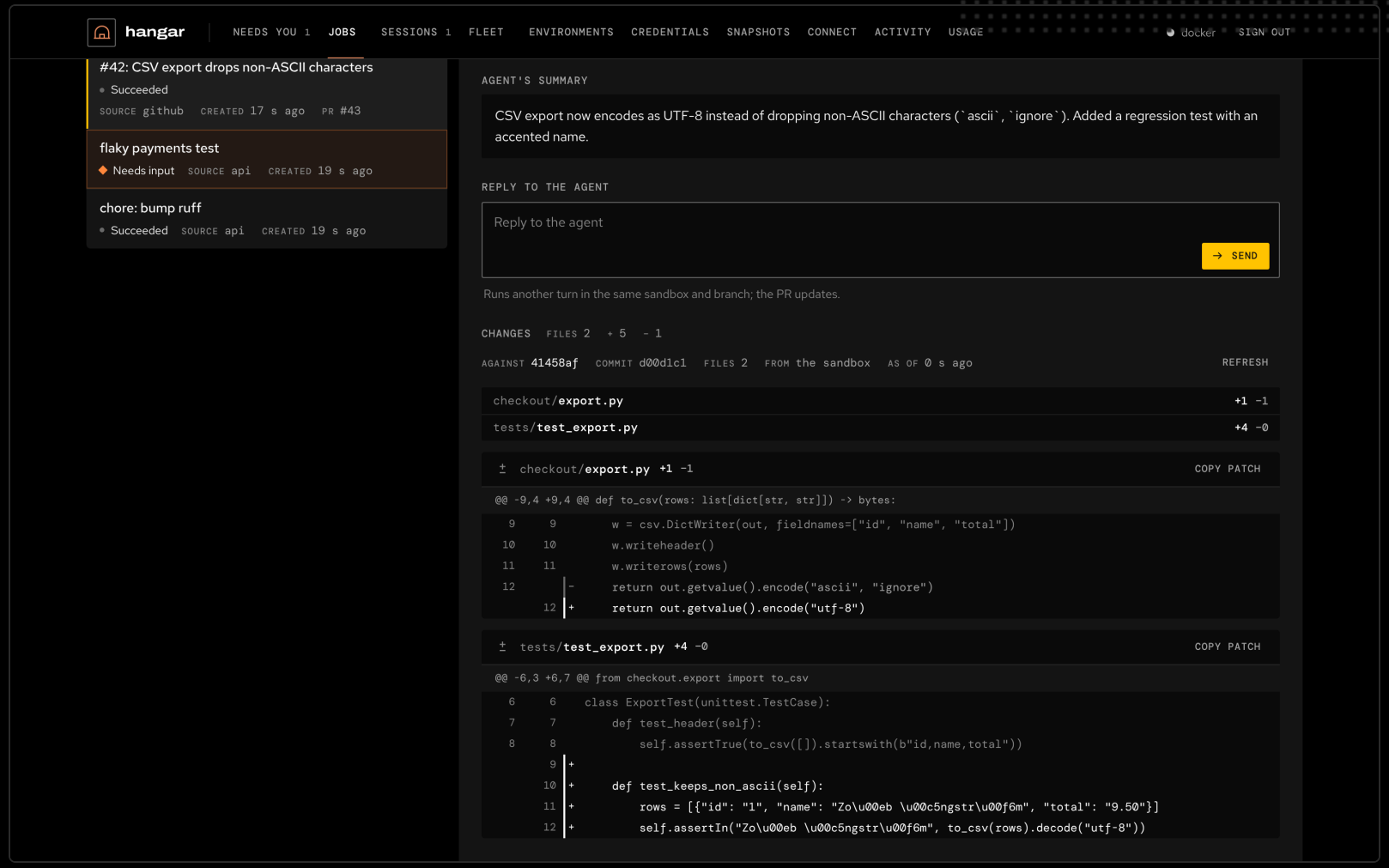
HANGAR · NEEDS YOU

NIGHT · 1440 × 900

Review the change, reply to the agent

A job's detail carries the agent's summary, a composer and the change as a greyscale diff. A comment on a line rides with your next message; the PR updates on the agent's next turn.

- **+5 −1** across 2 files, against the base commit, from the sandbox.
- Copy the patch, or open the agent's terminal and work beside it.
- Only a failing test lights a hunk; additions and deletions stay grey.



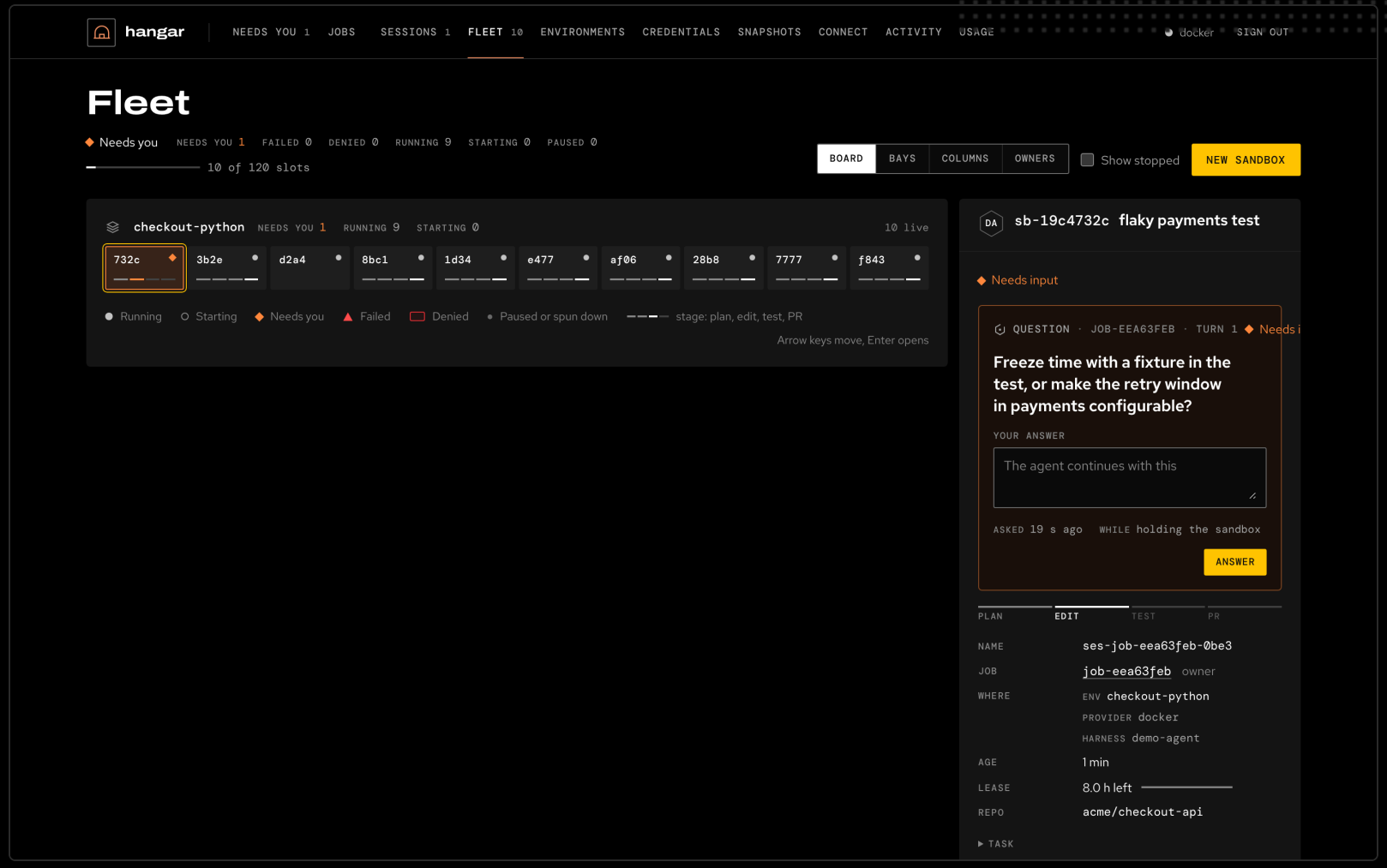
HANGAR · JOB #42, THE CHANGE

NIGHT · 1440 × 900

The whole fleet on one board

Board, Bays, Columns and Owners are four views of the same sandboxes and one selection. **10 of 120** slots in use; the one tile that needs a person is the only color on the page.

- Arrow keys move across the board; Enter opens the detail beside it.
- Each tile shows its stage: plan, edit, test, PR.
- Allow or deny a held step, or answer a question, without leaving the Fleet.

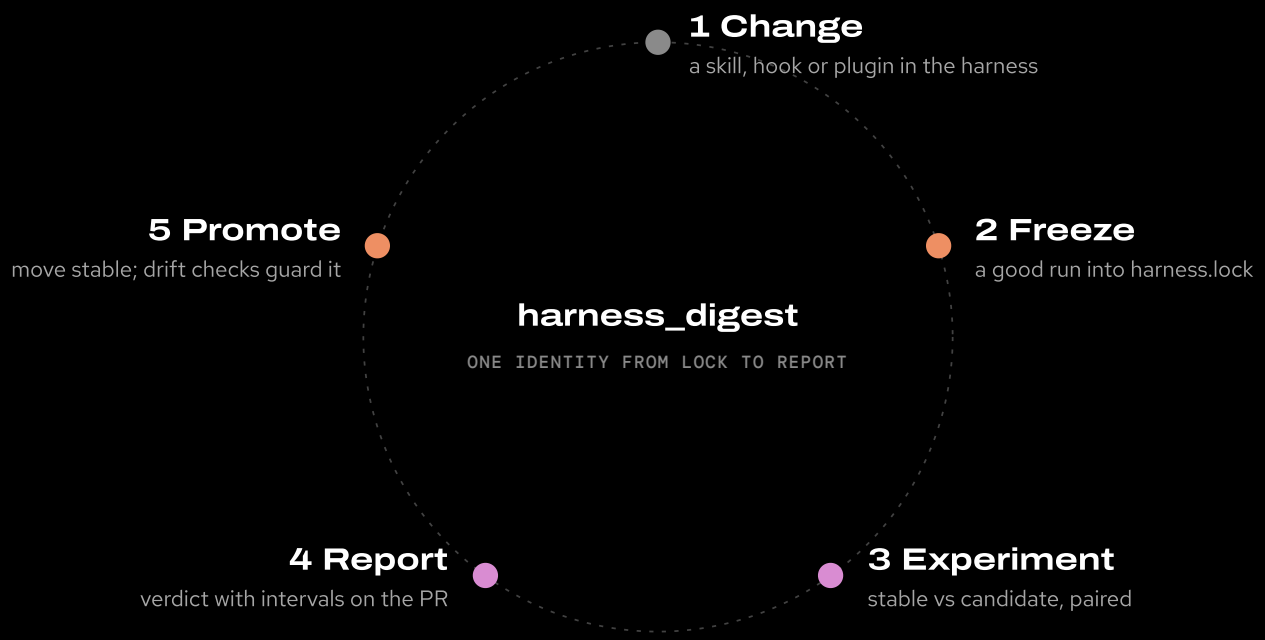


HANGAR · FLEET, BOARD VIEW

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CAPTURED FROM THE PRODUCT'S OWN DEMO BY SCRIPTS/CAPTURES.PY

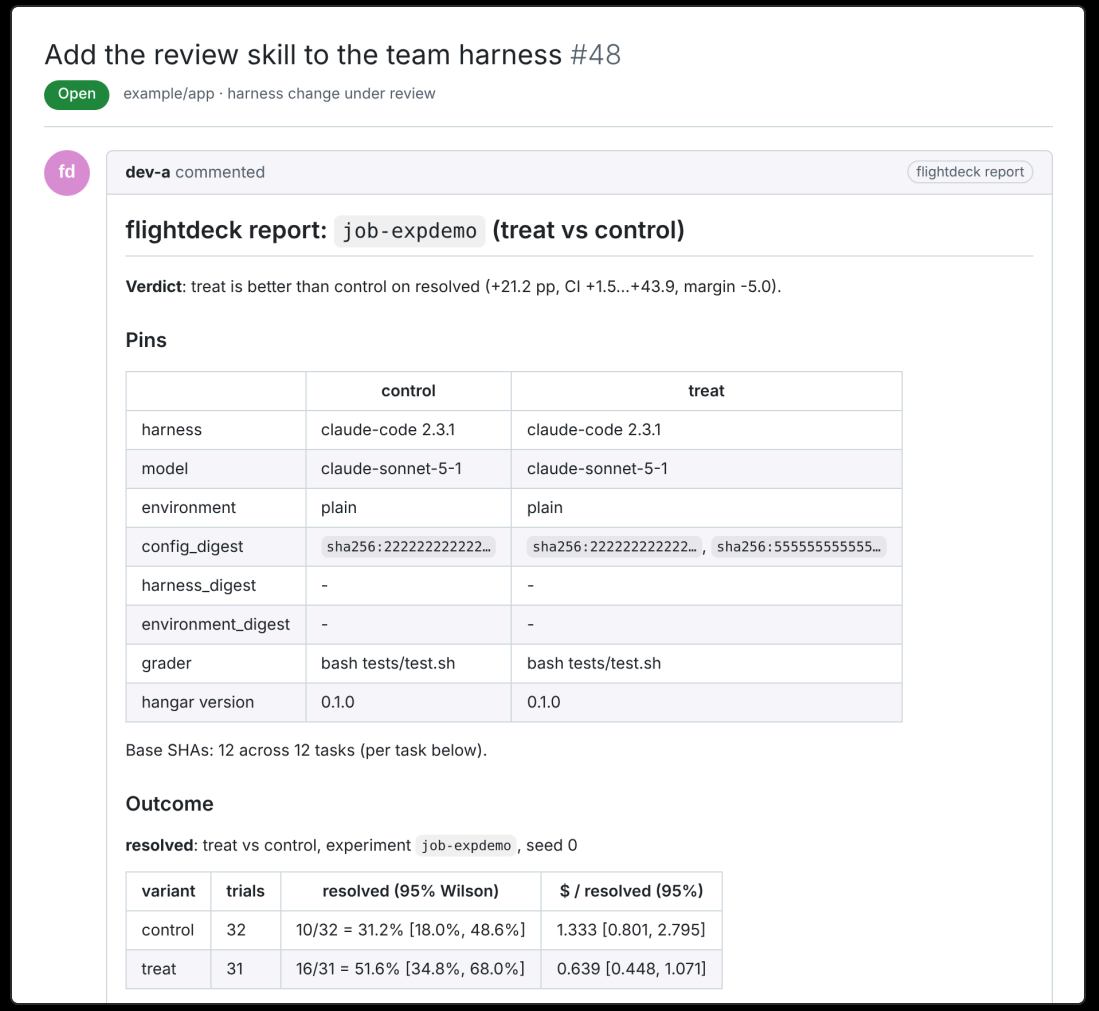
Change the harness on evidence, not impressions



The verdict lands on the harness PR

flightdeck posts one comment on the pull request that changes the harness, and updates it in place: the verdict, the pins of both variants and the outcome with its interval.

- **`treat is better than control`** on resolved: +21.2 pp, CI +1.5 to +43.9.
- Paired by task; cancelled, drifted and infrastructure failures are excluded and listed.
- The approver is never the author.



Add the review skill to the team harness #48

Open · example/app · harness change under review

fd **dev-a** commented — flightdeck report

flightdeck report: `job-expdemo` (treat vs control)

Verdict: treat is better than control on resolved (+21.2 pp, CI +1.5...+43.9, margin -5.0).

Pins

	control	treat
harness	claude-code 2.3.1	claude-code 2.3.1
model	claude-sonnet-5-1	claude-sonnet-5-1
environment	plain	plain
config_digest	`sha256:222222222222…`	`sha256:222222222222…` , `sha256:555555555555…`
harness_digest	-	-
environment_digest	-	-
grader	bash tests/test.sh	bash tests/test.sh
hangar version	0.1.0	0.1.0

Base SHAs: 12 across 12 tasks (per task below).

Outcome

resolved: treat vs control, experiment `job-expdemo`, seed 0

variant	trials	resolved (95% Wilson)	$ / resolved (95%)
control	32	10/32 = 31.2% [18.0%, 48.6%]	1.333 [0.801, 2.795]
treat	31	16/31 = 51.6% [34.8%, 68.0%]	0.639 [0.448, 1.071]

FLIGHTDECK · REPORT ON PR #48, FIXTURE EXPERIMENT

AS POSTED ON GITHUB

GOVERNANCE

Risky actions fail closed and leave a receipt

TAINT

Tracked

Reading a web page, a remote MCP server or a ticket marks the run.

PUSH

Fetch-only

Sandboxes can't push. hangar delivers one commit through its gateway.

DECISION POINT

Ask first

Release, landing, egress and secrets go to Tower: allow, deny or wait.

APPROVER

Never you

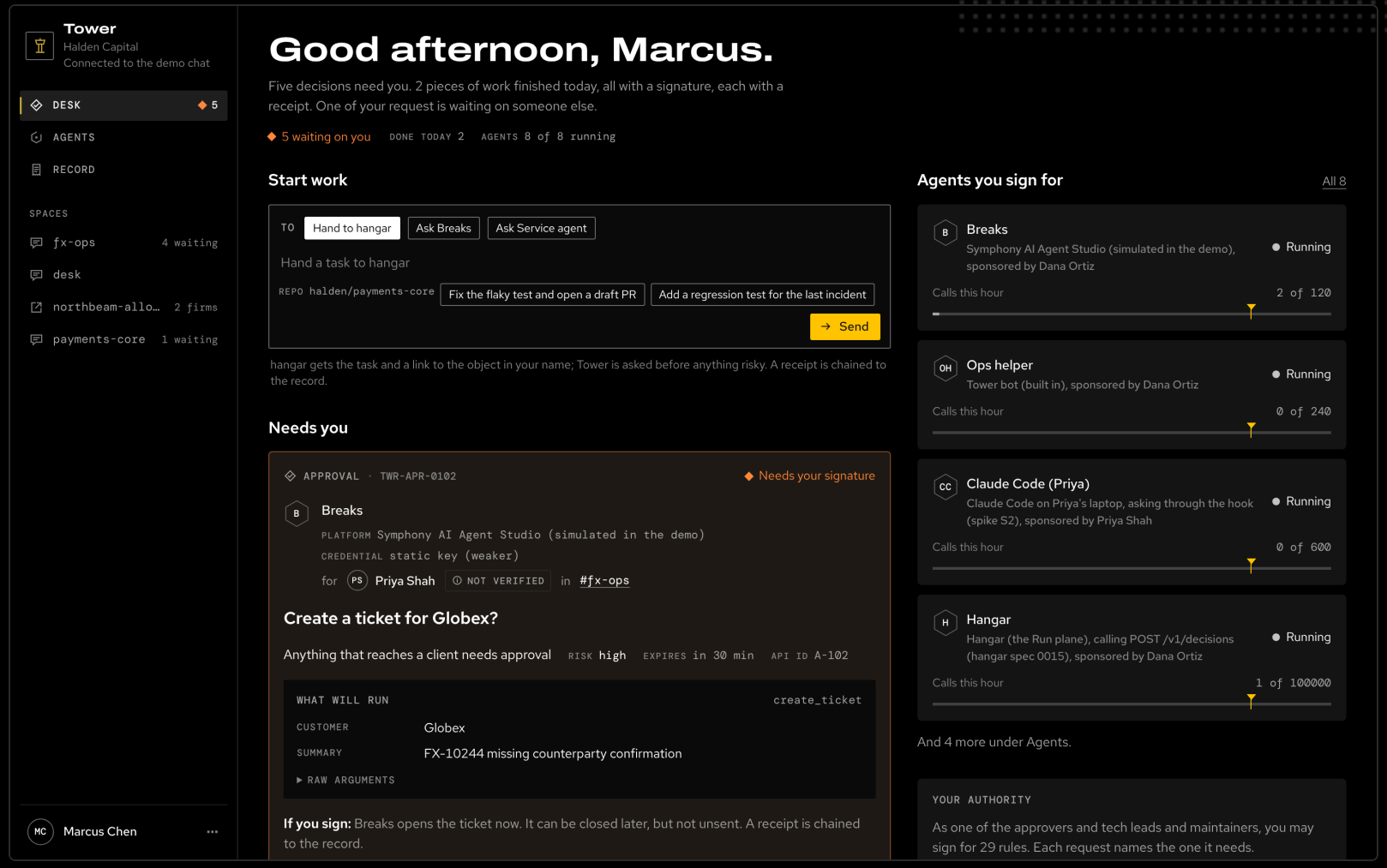
The requester never approves their own risky work. Every decision is signed.

ROLLOUT: OFF, THEN SHADOW, THEN ENFORCE, ONE DECISION KIND AT A TIME

The Desk: what needs you, what finished, who you sign for

Tower's home for a person. Start work from the composer, hand a task to hangar or ask an agent in its room, decide the cards that need your signature, and see the day's receipts.

- **5 waiting on you** · done today 2 · agents 8 of 8 running.
- Each card says what will run, why it asked and what signing does.
- Every decision files a receipt, chained into the record.



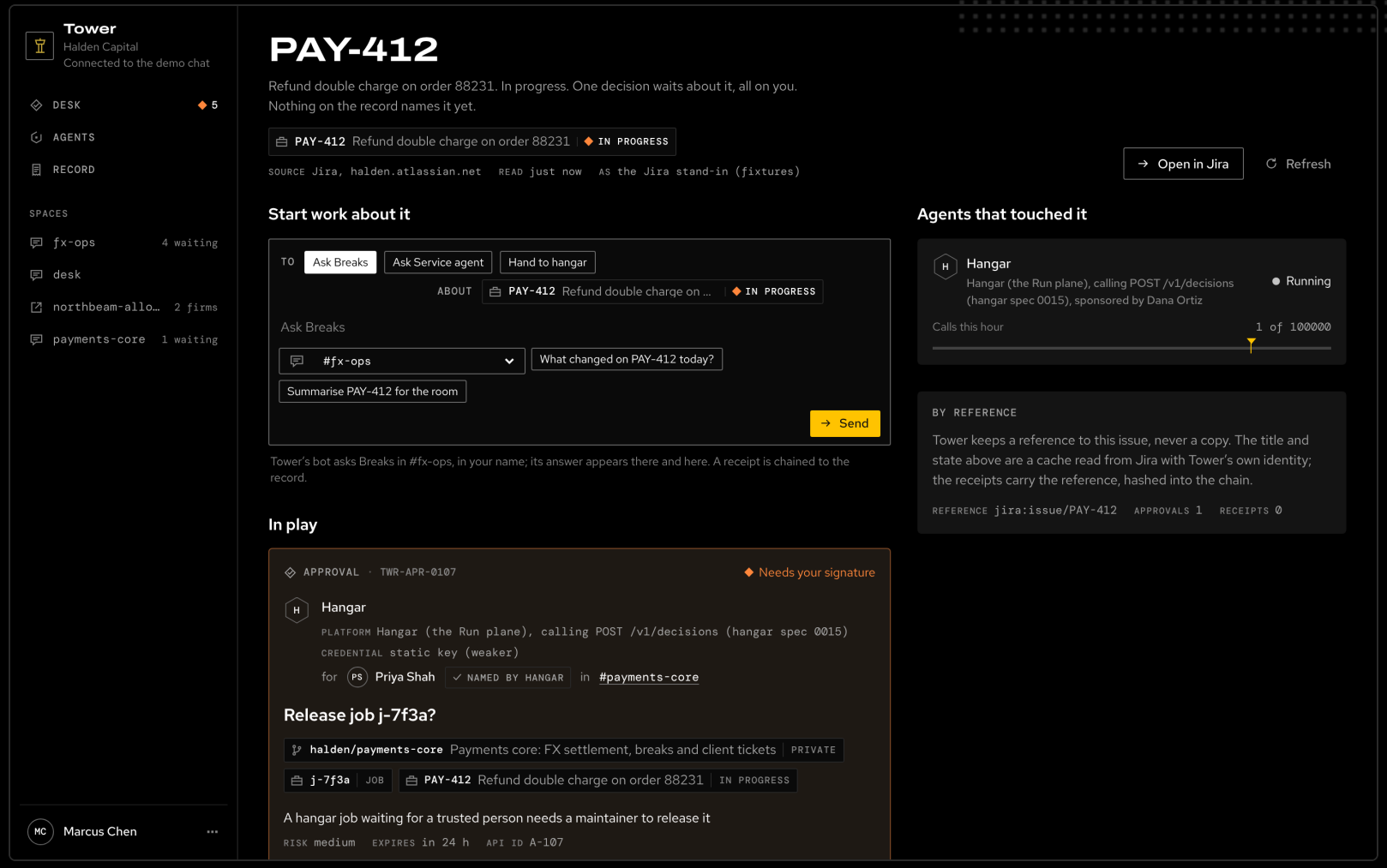
TOWER · THE DESK, AS AN APPROVER

NIGHT · 1440 × 900

Business objects by reference

A Jira issue, a pull request, a hangar job: each is a chip, and each chip opens a page with what touched it, what is in play and a composer to start work about it.

- Tower keeps a reference, never a copy; the title and state are read from the source.
- The receipt carries the reference, hashed into the chain.
- Ask Breaks in #fx-ops, in your name; the answer comes back here.



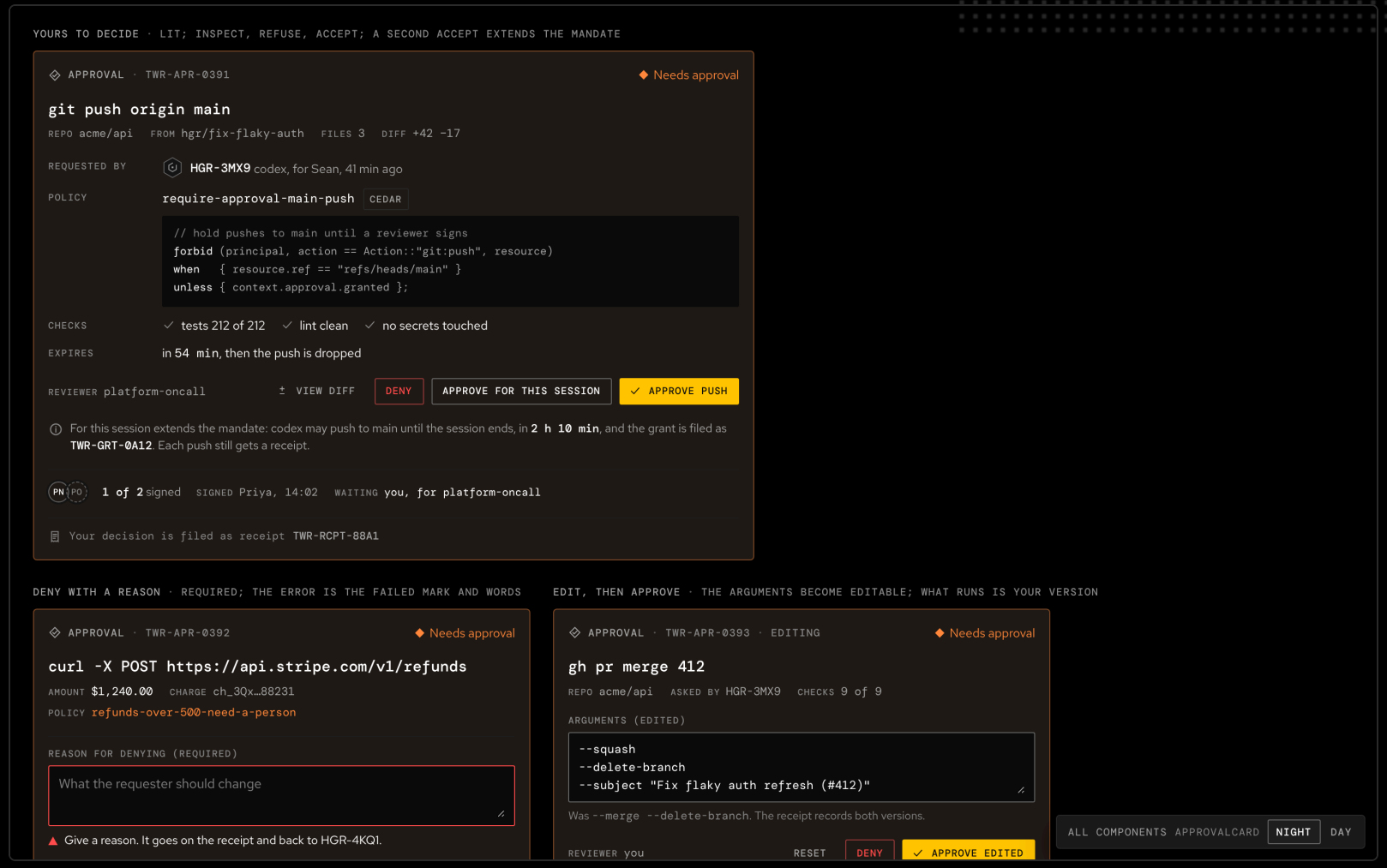
TOWER · OBJECT PAGE, JIRA:ISSUE/PAY-412

NIGHT · 1440 × 900

One design system, every screen

Airframe theme-v6 adds the parts a person needs to work with an agent: approval card, question, composer, diff, avatars, object chips, page header, empty state, toast, dialog and drawer. Every screen in this deck is built from it.

- Tokens and **af-** classes only; a test in each repo fails on a color literal.
- Night and Day, forced colors and reduced motion, in one stylesheet.
- 33 components in the gallery; the tool repos vendor a tagged release.



AIRFRAME · GALLERY, APPROVALCARD

NIGHT · 1440 × 900

CAPTURED FROM THE PRODUCT'S OWN DEMO BY SCRIPTS/CAPTURES.PY

— WHERE IT STANDS

Running and measured today, governed next

ON MAIN

Safe to run

- Sandboxes, jobs and sessions on GKE
- Intake posture, taint gates, sealed graders
- Manifests and traces with harness digests
- flightdeck data plane and statistics

ON MAIN

Measure everything

- Harness lock: freeze, check, thaw
- Run outbox and nightly compactor
- Reports, behavior specs, pinned trials
- Slack inbox, Tower's decision API

NEXT

Governed and decided

- Tower decisions in shadow, then enforce
- One real flightdeck decision
- Team harness, frozen from a good run
- Outbox and compactor on in the cluster